



IT6/L (8419) Fundamental of Database Systems

PAWS – Pet Adoption Transaction Management System

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I. Project Context

1.1 Background of the System

The Animal welfare and pet adoption in the Philippines, just like in any other countries, are mostly limited by manual, paper-based processes that slows down the services and has the risk of data loss. Studies on digital shelter systems show that moving to integrated web or mobile platforms can greatly improve record-keeping, coordination, and adoption outcomes.

There are several projects that demonstrate that manual processes struggle with rising rescue and adoption volumes, making it hard to manage records and applications accurately and on time.[1][2] Shelters frequently lack standardized, electronic data systems, which hampers understanding of animal populations and limits informed decision-making [3]. The Digital platforms for pet adoption and shelter management (web or mobile) have been developed to address these issues by centralizing animal profiles, health records, adoption requests, and communication among staff, veterinarians, and adopters [4].

To address these challenges, the PAWS: Pet Adoption Transaction Management System was developed as a web-based solution designed to digitalize and streamline the entire pet adoption lifecycle. The system was built using Laravel 12, Bootstrap 5, and MySQL within a XAMPP local development environment because PAWS integrates the key functionalities such as role-based access control, medical record management, payment tracking, and real-time notifications into a unified and efficient platform.

1.2 Description of Current Process

Before PAWS was developed, most animal shelters handled their day-to-day operations using a mix of physical logbooks, spreadsheets, and informal communication tools like social media and messaging apps. These methods worked to some extent, but they often caused confusion and delays.

Common practices included the following:

- Pet intake and profiles were written on paper forms, which made it hard to update or find a pet's health history later on.
- Adoption applications were submitted in person or through printed forms, with no proper system to track their status.

- Veterinary records, vaccination schedules, and health clearances were stored separately from adoption records, making information harder to organize.
- Adoption fees were collected and recorded not formally, with little to no tracking or documentation.
- Communication between staff, veterinarians, and applicants relied on personal messages, with no official system for updates or notifications.

1.3 Importance of the Project

The PAWS: Pet Adoption Management System is important because it helps improve how the animal shelters manage their daily operations. Instead of just relying on their scattered records and manual processes, the system brings everything together into one accessible web platform, which helps reduce paperwork and minimize errors.

It also introduces a more a more organized adoption process. Because of a dual-approval setup which involves both the veterinarian and the administrator, only pets that are healthy and cleared can be made available for adoption. This ensures better care for the animals and more confidence for adopters.

The system also provides different access levels for each type of user, so staff members can focus only on the tasks assigned to them. And lastly, the activities and payments are recorded within the system, making it easier to track transactions and maintain transparency.

II. Problem of the Statement

The following problems were identified in the current manual processes of animal shelters and welfare organizations:

1. Lack of a centralized database for pet records, resulting in incomplete and inconsistent health histories that cannot be easily retrieved or shared among shelter staff and veterinarians.
2. The Absence of a structured adoption workflow, causing delays and inconsistencies in processing adoption applications, conducting reviews, and communicating decisions to applicants.

3. No formal veterinary oversight mechanism integrated into the adoption process, making it difficult to ensure that all pets listed for adoption have passed health evaluations.
4. Inadequate financial recordkeeping for adoption fees and donations, with no reference numbers, payment methods tracking, or audit trails.
5. No role-based access control, meaning that sensitive records and administrative functions are accessible to all staff without appropriate restrictions.
6. Poor communication infrastructure, leading to delayed or missed notifications between staff, veterinarians, and adopters regarding application updates and pet availability.

III. Proposed Solution

- Web-based, role-based system in order to improve shelter operations
- Have a centralized MySQL database for pet records, medical data, applications, payments, and users
- Structured adoption process with clear stages and workflow
- Veterinarian role for health checks and adoption clearance
- Payment module with multiple methods and proper tracking
- Role-based access for Admin, Staff, Veterinarian, and Adopter
- Built-in notifications and activity logs for updates and accountability

IV. Objective of the Study

General Objective

To develop a web-based PAWS: Pet Adoption Transaction Management System that simplifies pet adoption, health tracking, and shelter management.

Specific Objectives

- Implement role-based access for Admin, Staff, Veterinarian, and Adopter
- Create a system to manage pet profiles and health records
- Enable veterinarians to record check-ups and approve pets for adoption
- Develop a structured adoption process with status tracking
- Add a payment system with proper recording.
- Provide notifications for important updates
- Track system activities for accountability
- Offer dashboards with summaries for better decision making.

V. Scopes and Limitations

The PAWS system covers the key shelter operations such as the user management with a role-based access, pet management from intake to adoption, adoption application processing, veterinary and vaccination records, payment tracking, notifications, activity logs, and basic reporting through dashboards. However, it has some limitations like it does not support online payment integration, real-time messaging, or email notifications, and is designed for single shelter use only. In addition, it is a web-based system without a mobile app, and only registered users can browse available pets.

VI. System Analysis and Design

6.1 Business Rules

- A user must have at least one role to access the system
- Only Admin can assign, update, or remove user roles
- A user can have multiple roles with combined permissions
- Suspended users are not allowed to log in
- A pet must be examined by a veterinarian before approval
- Only Admin can approve pets for public listing
- Only approved and available pets are visible to adopters
- Pets use soft delete to keep historical records
- An adopter can submit one active application per pet
- Applications follow a fixed status workflow
- Only Staff and Admin can review and manage applications
- Only Admin can approve or reject applications
- Approved applications automatically mark the pet as adopted
- Only Veterinarians can manage medical and vaccination records
- Vaccination records are tied to the assigned Veterinarian
- Payments must be linked to an adoption application
- Only Staff and Admin can manage payment records
- Each payment has a unique reference number and status
- The system generates notifications for important events
- All major actions are recorded in activity logs for tracking

6.2 Conceptual Design (ERD)

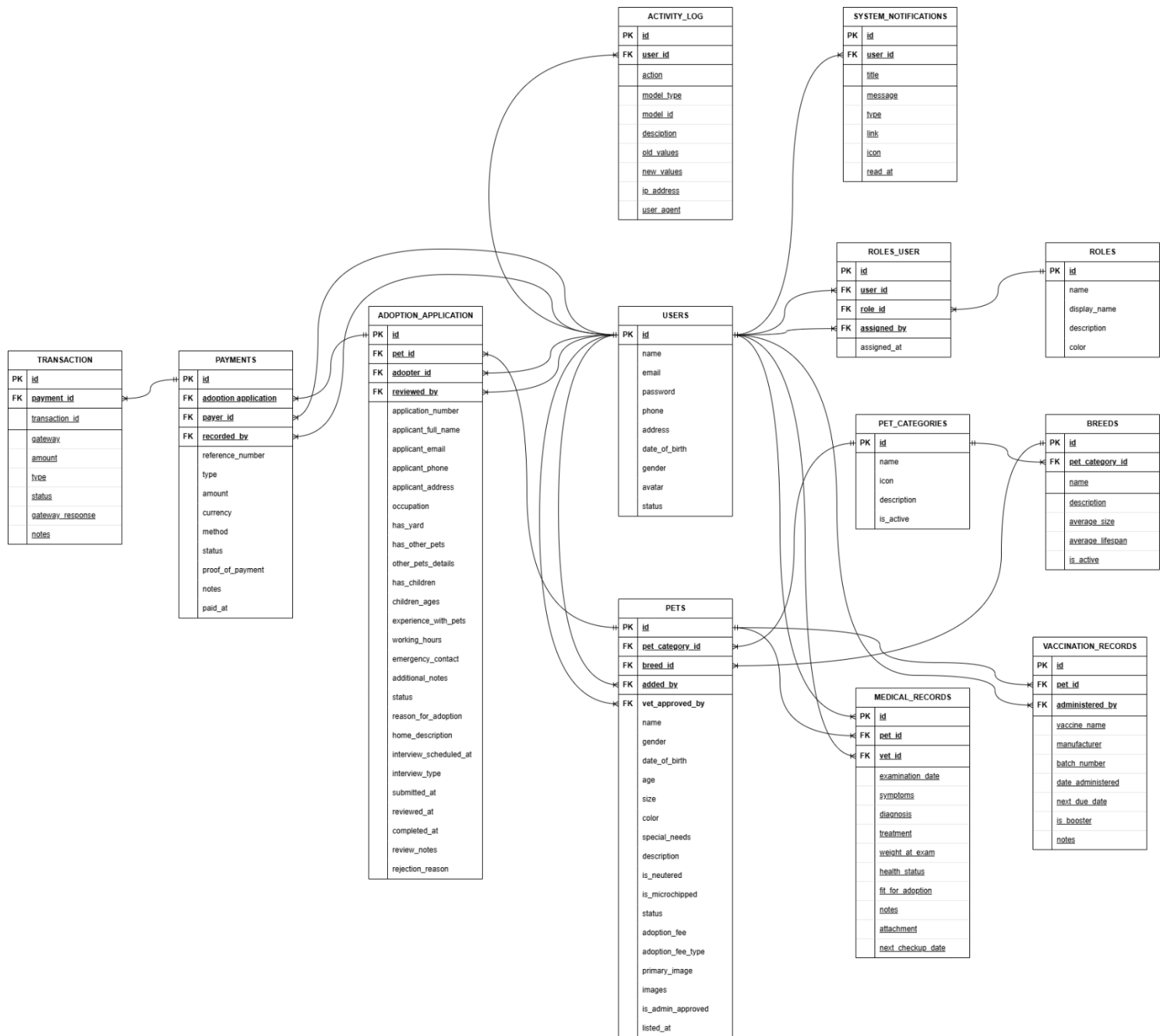


Figure 1. Full Blown ERD

6.3 Explanation of entities, relationships, and processes

A. MAPPING ENTITIES TO RELATION

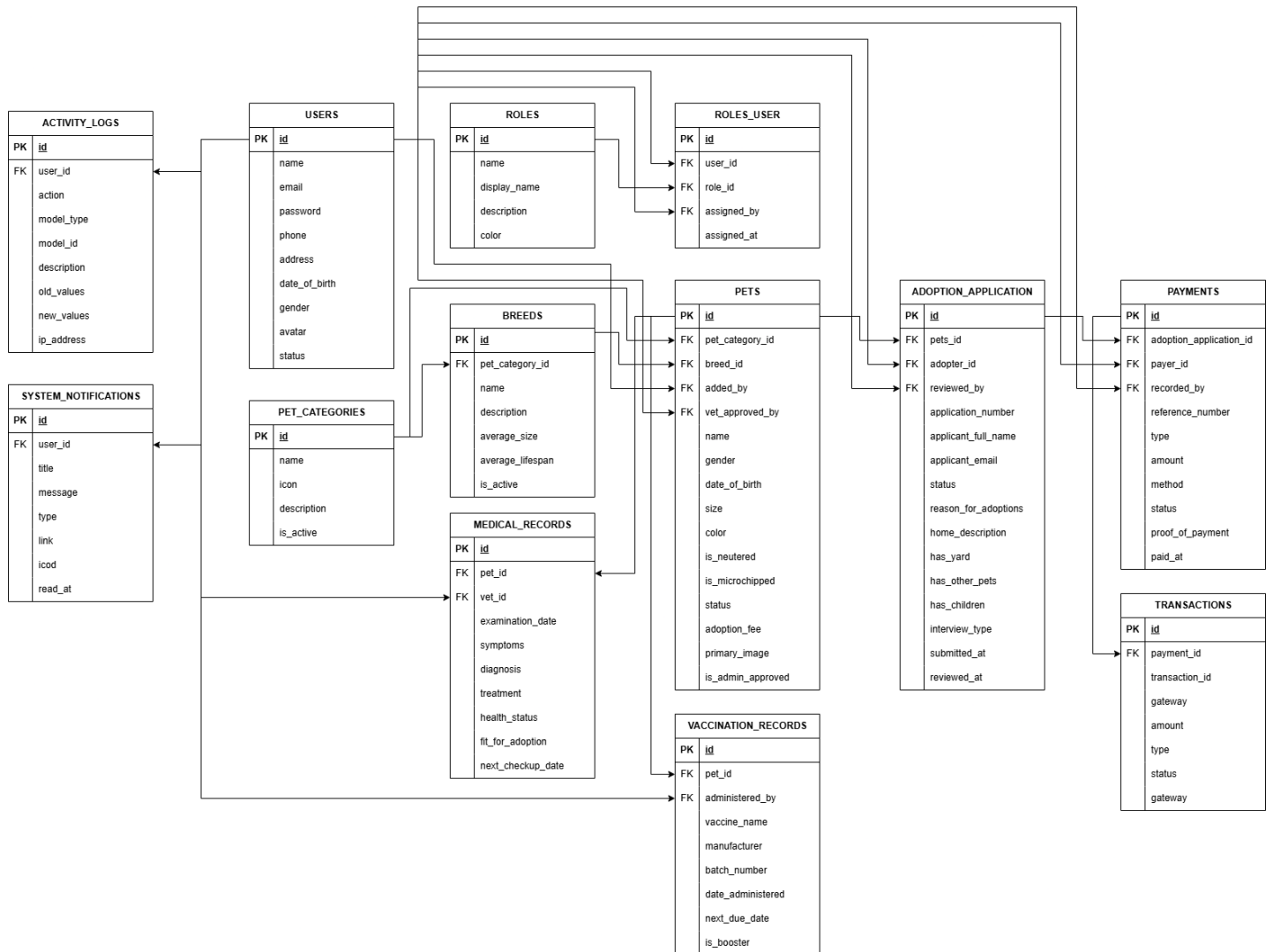


Figure 2. Entity Mapping

B. NORMALIZATION

Zero NF

adopter_name	adopter_email	adopter_phone	pet_name	pet_category	breed_name	pet_status	vet_name	diagnosis	vaccine_name	date_administered	application_status	payment_amount	payment_method	staff_name	staff_role	activity_action
Juan Dela Cruz	juan@email.com	092171234567	Buddy	Dog	Aspin	Available	Dr. Santos	Healthy	Rabies	2024-01-10	Approved	500.00	Value 5	Maria Reyes	Staff	createPet
Juan Dela Cruz	juan@email.com	092171234567	Buddy	Dog	Aspin	Available	Dr. Santos	Healthy	Distemper	2024-01-15	Approved	500.00	Value 8	Maria Reyes	Staff	approved_application

Figure 3. Zero NF

First NF - No Repeating Groups

Adoption Record(1NF)	
PK	record_id
	adopter_name
	adopter_email
	adopter_phone
	pet_name
	pet_category
	breed_name
	vet_name
	diagnosis
	vaccine_name
	date_administered
	application_status
	payment_amount
	payment_method
	staff_name
	staff_role
	activity_action

Figure 4. First NF

Second NF - No Partial Dependencies

Users	
PK	id
	name
	email
	password
	phone
	address
	gender
	status

Roles	
PK	id
	name
	display_name
	description
	color

Pets	
PK	id
FK	category_id
FK	breed_id
FK	added_by
	name
	gender
	status
	adoption_fee

Pet Categories	
PK	id
	name
	icon
	description
	is_active

Breeds	
PK	id
FK	pet_category_id
	name
	average_size
	average_lifespan

Adoption Application	
PK	id
FK	pet_id
FK	adopter_id
	application_number
	status
	reason_for_adoption
	submitted_at

Medical Records	
PK	id
FK	pet_id
FK	vet_id
	examination_date
	diagnosis
	health_status
	fit_for_adoption

Vaccination Records	
PK	id
FK	pet_id
FK	administered_by
	examination_date
	date_administered
	next_due_date

Vaccination Records	
PK	id
FK	application_id
FK	payer_id
	reference_number
	amount
	method
	status

Figure 5. Second NF

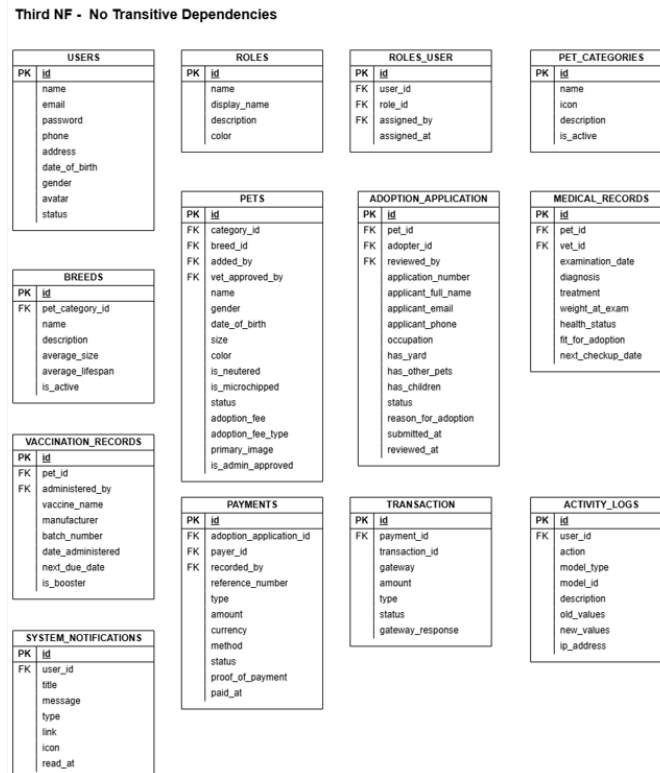


Figure 6. Third NF

The PAWS - Pet Adoption Transaction Management System database is consisting of 13 core tables that is organized into five functional groups:

Users and Access Control

The USERS table stores all registered accounts regardless of role. The ROLES table defines the four system roles: Administrator, Staff, Veterinarian, and Adopter. The ROLE_USER junction table is the one that implements the many-to-many relationship between users and roles, storing the assignment timestamp and the identity of the assigning user. This structure allows one user to hold multiple roles simultaneously.

Pet Catalog

The PET_CATEGORIES table and BREEDS table form a two-level taxonomy for classifying pets. PET_CATEGORIES holds top-level types such as Dog, Cat, and Bird, while BREEDS

holds specific breeds linked to each category. The PETS table is the central entity of the system, referencing both its category and breed through foreign keys, and linking to the users who added and approved it.

Adoption Workflow

The ADOPTION_APPLICATIONS table is the one that captures the full adoption form submitted by the adopter, which includes their personal information, home description, lifestyle details, and their interview scheduling data. It also references the pet that are being applied for, the adopter, and the reviewing staff member. The PAYMENTS table is linked to an approved application and records the financial transaction, while the TRANSACTIONS table logs individual gateway or payment events for each payment record.

Medical Management

The MEDICAL_RECORDS table stores health assessments created by veterinarians for each pet, including examination findings, treatment plans, and the fit-for-adoption flag. The VACCINATION_RECORDS table logs individual vaccine administrations linked to a pet and the user who administered the vaccine.

Audit and Notifications

The ACTIVITY_LOGS table uses a polymorphic structure with model_type and model_id columns, allowing it to reference any table in the system without requiring separate log tables. This records all significant actions across the application. The SYSTEM_NOTIFICATIONS table stores per-user alerts generated by system events, with a read_at timestamp enabling unread notification tracking.

VII. Database Design

7.1 List of Tables

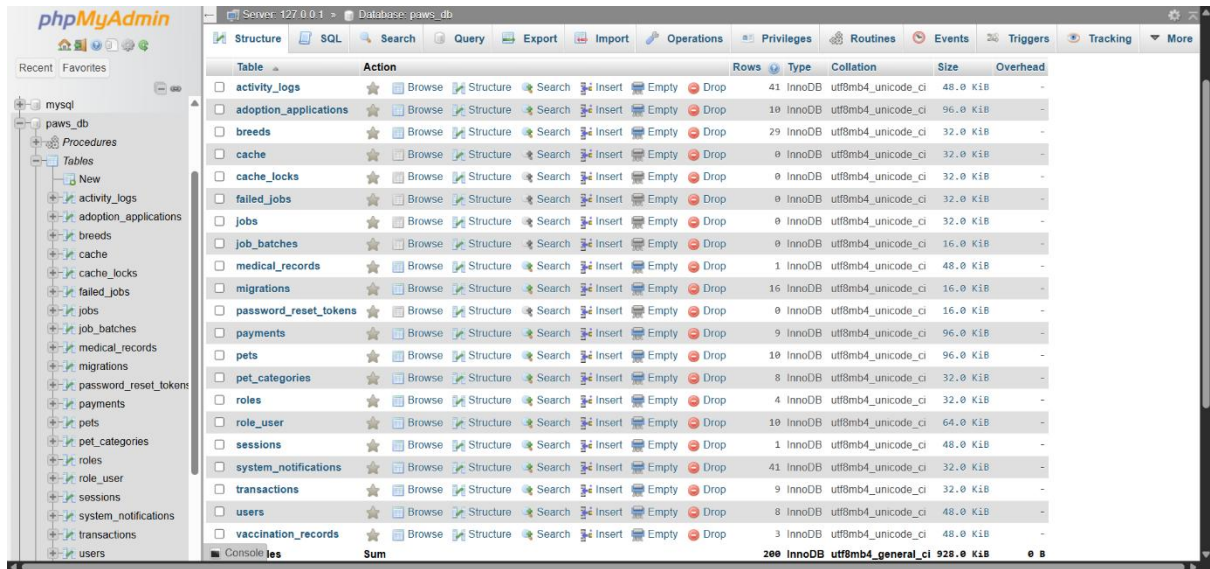


Table	Action	Rows	Type	Collation	Size	Overhead
activity_logs	Browse Structure Search Insert Empty Drop	41	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
adoption_applications	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_unicode_ci	96.0 KiB	-
breeds	Browse Structure Search Insert Empty Drop	29	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache_locks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
failed_jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
job_batches	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
medical_records	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
migrations	Browse Structure Search Insert Empty Drop	16	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
password_reset_tokens	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
payments	Browse Structure Search Insert Empty Drop	9	InnoDB	utf8mb4_unicode_ci	96.0 KiB	-
pets	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_unicode_ci	96.0 KiB	-
pet_categories	Browse Structure Search Insert Empty Drop	8	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
roles	Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
role_user	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
sessions	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
system_notifications	Browse Structure Search Insert Empty Drop	41	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
transactions	Browse Structure Search Insert Empty Drop	9	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
users	Browse Structure Search Insert Empty Drop	8	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
vaccination_records	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
Sum		200	InnoDB	utf8mb4_general_ci	928.0 KiB	0 B

Note: The Tables that are highlighted in the list such as the cache, sessions, jobs, migrations, and the password_reset_tokens are Laravel framework system tables and are not part of the core application design.

7.2 Fields and Data Types

Users Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 email 📧	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 phone	varchar(20)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 address	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 date_of_birth	date			Yes	NULL			Change Drop More
☐	7 gender	enum('male', 'female', 'other')	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 avatar	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9 status	enum('active', 'inactive', 'suspended')	utf8mb4_unicode_ci		No	active			Change Drop More
☐	10 last_login_at	timestamp			Yes	NULL			Change Drop More
☐	11 notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	12 email_verified_at	timestamp			Yes	NULL			Change Drop More
☐	13 password	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	14 remember_token	varchar(100)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	15 created_at	timestamp			Yes	NULL			Change Drop More
☐	16 updated_at	timestamp			Yes	NULL			Change Drop More

Roles Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 name 📧	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 display_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 color	varchar(20)	utf8mb4_unicode_ci		No	secondary			Change Drop More
☐	6 created_at	timestamp			Yes	NULL			Change Drop More
☐	7 updated_at	timestamp			Yes	NULL			Change Drop More

Role User Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id 📧	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 role_id 📧	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 assigned_at	timestamp			No	current_timestamp()			Change Drop More
☐	5 assigned_by 📧	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	6 created_at	timestamp			Yes	NULL			Change Drop More
☐	7 updated_at	timestamp			Yes	NULL			Change Drop More

Pet Categories Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name 🔑	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 3	icon	varchar(10)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 4	description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 5	is_active	tinyint(1)			No	1			Change Drop More
☐ 6	created_at	timestamp			Yes	NULL			Change Drop More
☐ 7	updated_at	timestamp			Yes	NULL			Change Drop More
☐ 8	deleted_at	timestamp			Yes	NULL			Change Drop More

Breeds Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐ 2	pet_category_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐ 3	name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐ 4	description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 5	average_size	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 6	average_lifespan	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐ 7	is_active	tinyint(1)			No	1			Change Drop More
☐ 8	created_at	timestamp			Yes	NULL			Change Drop More
☐ 9	updated_at	timestamp			Yes	NULL			Change Drop More
☐ 10	deleted_at	timestamp			Yes	NULL			Change Drop More

Pets Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 pet_category_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 breed_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	5 gender	enum('male', 'female', 'unknown')	utf8mb4_unicode_ci		No	unknown			Change Drop More
☐	6 date_of_birth	date			Yes	NULL			Change Drop More
☐	7 weight	decimal(5,2)			Yes	NULL			Change Drop More
☐	8 size	enum('tiny', 'small', 'medium', 'large', 'extra_la...)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9 color	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 special_needs	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	12 is_vaccinated	tinyint(1)			No	0			Change Drop More
☐	13 is_neutered	tinyint(1)			No	0			Change Drop More
☐	14 is_microchipped	tinyint(1)			No	0			Change Drop More
☐	15 status 🔑	enum('available', 'pending', 'adopted', 'under_tre...)	utf8mb4_unicode_ci		No	available			Change Drop More
☐	16 adoption_fee_type	enum('fixed', 'donation', 'free')	utf8mb4_unicode_ci		No	fixed			Change Drop More
☐	17 adoption_fee	decimal(8,2)			No	0.00			Change Drop More
☐	18 primary_image	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	19 images	longtext	utf8mb4_bin		Yes	NULL			Change Drop More
☐	20 added_by 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	21 vet_approved_by 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	22 is_vet_approved	tinyint(1)			No	0			Change Drop More
☐	23 is_admin_approved 🔑	tinyint(1)			No	0			Change Drop More
☐	24 listed_at	timestamp			Yes	NULL			Change Drop More
☐	25 created_at	timestamp			Yes	NULL			Change Drop More
☐	26 updated_at	timestamp			Yes	NULL			Change Drop More
☐	27 deleted_at	timestamp			Yes	NULL			Change Drop More

Adoption Application Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 application_number 🔑	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 pet_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 adopter_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	5 reviewed_by 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	6 applicant_full_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	7 applicant_email	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	8 applicant_phone	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	9 applicant_address	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐	10 housing_type	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	11 has_yard	tinyint(1)			No	0			Change Drop More
☐	12 has_other_pets	tinyint(1)			No	0			Change Drop More
☐	13 other_pets_details	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	14 has_children	tinyint(1)			No	0			Change Drop More
☐	15 children_ages	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	16 reason_for_adopting	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐	17 experience_with_pets	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	18 occupation	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	19 working_hours	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	20 emergency_contact	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	21 additional_notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	22 status 🔑	enum('draft','submitted','under_review','interv...)	utf8mb4_unicode_ci		No	submitted			Change Drop More
☐	23 submitted_at	timestamp			Yes	NULL			Change Drop More
☐	24 reviewed_at	timestamp			Yes	NULL			Change Drop More
☐	25 interview_scheduled_at	timestamp			Yes	NULL			Change Drop More
☐	26 completed_at	timestamp			Yes	NULL			Change Drop More
☐	27 review_notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	28 rejection_reason	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	29 created_at	timestamp			Yes	NULL			Change Drop More
☐	30 updated_at	timestamp			Yes	NULL			Change Drop More
☐	31 deleted_at	timestamp			Yes	NULL			Change Drop More

Medical Records Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 pet_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 vet_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 examination_date	date			No	None			Change Drop More
☐	5 diagnosis	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 symptoms	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 treatment	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 medications	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9 weight_at_exam	decimal(5,2)			Yes	NULL			Change Drop More
☐	10 health_status	enum('excellent','good','fair','poor','critica...)	utf8mb4_unicode_ci		No	good			Change Drop More
☐	11 fit_for_adoption	tinyint(1)			No	1			Change Drop More
☐	12 notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	13 attachment	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	14 next_checkup_date	date			Yes	NULL			Change Drop More
☐	15 created_at	timestamp			Yes	NULL			Change Drop More
☐	16 updated_at	timestamp			Yes	NULL			Change Drop More
☐	17 deleted_at	timestamp			Yes	NULL			Change Drop More

Vaccination Records Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 pet_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 administered_by	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	4 vaccine_name	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 manufacturer	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 batch_number	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 date_administered	date			No	None			Change Drop More
☐	8 next_due_date	date			Yes	NULL			Change Drop More
☐	9 is_booster	tinyint(1)			No	0			Change Drop More
☐	10 notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 created_at	timestamp			Yes	NULL			Change Drop More
☐	12 updated_at	timestamp			Yes	NULL			Change Drop More

Payments Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 reference_number	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	3 adoption_application_id	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	4 payer_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	5 recorded_by	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	6 type	enum('adoption_fee', 'donation', 'medical_fee', 'o...)	utf8mb4_unicode_ci		No	adoption_fee			Change Drop More
☐	7 amount	decimal(10,2)			No	None			Change Drop More
☐	8 currency	varchar(3)	utf8mb4_unicode_ci		No	PHP			Change Drop More
☐	9 method	enum('cash', 'bank_transfer', 'gcash', 'maya', 'cr...)	utf8mb4_unicode_ci		No	cash			Change Drop More
☐	10 status	enum('pending', 'completed', 'failed', 'refunded', ...)	utf8mb4_unicode_ci		No	pending			Change Drop More
☐	11 proof_of_payment	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	12 notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	13 paid_at	timestamp			Yes	NULL			Change Drop More
☐	14 created_at	timestamp			Yes	NULL			Change Drop More
☐	15 updated_at	timestamp			Yes	NULL			Change Drop More
☐	16 deleted_at	timestamp			Yes	NULL			Change Drop More

Transaction Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 payment_id	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 transaction_id	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	4 gateway	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 amount	decimal(10,2)			No	None			Change Drop More
☐	6 type	enum('credit', 'debit', 'refund')	utf8mb4_unicode_ci		No	credit			Change Drop More
☐	7 status	enum('success', 'failed', 'pending')	utf8mb4_unicode_ci		No	pending			Change Drop More
☐	8 gateway_response	longtext	utf8mb4_bin		Yes	NULL			Change Drop More
☐	9 notes	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10 created_at	timestamp			Yes	NULL			Change Drop More
☐	11 updated_at	timestamp			Yes	NULL			Change Drop More

Activity Logs Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	3 action	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 model_type 🔑	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 model_id 🔑	bigint(20)		UNSIGNED	Yes	NULL			Change Drop More
☐	6 description	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 old_values	longtext	utf8mb4_bin		Yes	NULL			Change Drop More
☐	8 new_values	longtext	utf8mb4_bin		Yes	NULL			Change Drop More
☐	9 ip_address	varchar(45)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10 user_agent	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11 created_at	timestamp			Yes	NULL			Change Drop More
☐	12 updated_at	timestamp			Yes	NULL			Change Drop More

System Notification Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id 🔑	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id 🔑	bigint(20)		UNSIGNED	No	None			Change Drop More
☐	3 title	varchar(255)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 message	text	utf8mb4_unicode_ci		No	None			Change Drop More
☐	5 type	varchar(255)	utf8mb4_unicode_ci		No	info			Change Drop More
☐	6 link	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	7 icon	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	8 read_at 🔑	timestamp			Yes	NULL			Change Drop More
☐	9 created_at	timestamp			Yes	NULL			Change Drop More
☐	10 updated_at	timestamp			Yes	NULL			Change Drop More

7.3 Constraints (Primary Key, Foreign Key)

The following constraints are applied across the PAWS database:

Constraint	Tables Applied	Description
PRIMARY KEY (PK)	All 13 tables	Each table has an auto-incrementing bigint id as its primary key
FOREIGN KEY (FK)	ROLE_USER, PETS, ADOPTION_APPLICATIONS, MEDICAL_RECORDS, VACCINATION_RECORDS, PAYMENTS, TRANSACTIONS, ACTIVITY_LOGS, SYSTEM_NOTIFICATIONS	Foreign keys enforce referential integrity between related tables
UNIQUE	USERS.email, ROLE_USER(user_id, role_id)	Prevents duplicate email addresses and duplicate role assignments
NOT NULL	All PK and FK columns, required form fields	Ensures critical fields always contain a value
SOFT DELETE	PETS, BREEDS, PET_CATEGORIES, ADOPTION_APPLICATIONS, MEDICAL_RECORDS, PAYMENTS	Adds deleted_at timestamp instead of permanently removing records
CASCADE DELETE	PETS → MEDICAL_RECORDS, VACCINATION_RECORDS	Deleting a pet automatically removes its associated health records
SET NULL	ACTIVITY_LOGS.user_id	If a user is deleted, activity logs retain the record but set user_id to null
ENUM	gender, status, type, method, health_status, adoption_fee_type	Restricts column values to a predefined set of valid options

Constraint Tables

Table	Column	Constraint	Reference
users	id	PRIMARY KEY	-
roles	id	PRIMARY KEY	-
role_user	id	PRIMARY KEY	-
role_user	user_id	FOREIGN KEY	users.id
role_user	role_id	FOREIGN KEY	roles.id
role_user	assigned_by	FOREIGN KEY	users.id
pet_categories	id	PRIMARY KEY	-
breeds	id	PRIMARY KEY	-
breeds	pet_category_id	FOREIGN KEY	pet_categories.id
pets	id	PRIMARY KEY	-
pets	pet_category_id	FOREIGN KEY	pet_categories.id
pets	breed_id	FOREIGN KEY	breeds.id
pets	added_by	FOREIGN KEY	users.id
pets	vet_approved_by	FOREIGN KEY	users.id
adoption_applications	id	PRIMARY KEY	-
adoption_applications	pet_id	FOREIGN KEY	pets.id
adoption_applications	adopter_id	FOREIGN KEY	users.id
adoption_applications	reviewed_by	FOREIGN KEY	users.id
medical_records	id	PRIMARY KEY	-
medical_records	pet_id	FOREIGN KEY	pets.id
medical_records	vet_id	FOREIGN KEY	users.id
vaccination_records	id	PRIMARY KEY	-
vaccination_records	pet_id	FOREIGN KEY	pets.id

vaccination_records	administered_by	FOREIGN KEY	users.id
payments	id	PRIMARY KEY	-
payments	adoption_application_id	FOREIGN KEY	adoption_applications.id
payments	payer_id	FOREIGN KEY	users.id
payments	recorded_by	FOREIGN KEY	users.id
transactions	id	PRIMARY KEY	-
transactions	payment_id	FOREIGN KEY	payments.id
activity_logs	id	PRIMARY KEY	-
activity_logs	user_id	FOREIGN KEY	users.id
system_notifications	id	PRIMARY KEY	-
system_notifications	user_id	FOREIGN KEY	users.id

Foreign Key Constraints (phpMyAdmin)

Role User Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	role_user_assigned_by_for ON DELETE SET NULL ON UPDATE RESTRICT	assigned_by	paws_db	users	id
Drop	role_user_role_id_foreign ON DELETE CASCADE ON UPDATE RESTRICT	role_id	paws_db	roles	id
Drop	role_user_user_id_foreign ON DELETE CASCADE ON UPDATE RESTRICT	user_id	paws_db	users	id
	Constraint name ON DELETE RESTRICT ON UPDATE RESTRICT		paws_db		

+ Add constraint

Breeds Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	breeds_pet_category_id_fo ON DELETE CASCADE ON UPDATE RESTRICT	pet_category_id	paws_db	pet_categories	id
	Constraint name ON DELETE RESTRICT ON UPDATE RESTRICT		paws_db		

+ Add constraint

Pet Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	pets_added_by_foreign ON DELETE RESTRICT ON UPDATE RESTRICT	added_by + Add column	paws_db	users	id
Drop	pets_breed_id_foreign ON DELETE SET NULL ON UPDATE RESTRICT	breed_id + Add column	paws_db	breeds	id
Drop	pets_pet_category_id_foreign ON DELETE RESTRICT ON UPDATE RESTRICT	pet_category_id + Add column	paws_db	pet_categories	id
Drop	pets_vet_approved_by_foreign ON DELETE SET NULL ON UPDATE RESTRICT	vet_approved_by + Add column	paws_db	users	id
	Constraint name ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column	paws_db		
+ Add constraint					

Adoption Application Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	adoption_applications_adopter_id_foreign ON DELETE RESTRICT ON UPDATE RESTRICT	adopter_id + Add column	paws_db	users	id
Drop	adoption_applications_pet_id_foreign ON DELETE RESTRICT ON UPDATE RESTRICT	pet_id + Add column	paws_db	pets	id
Drop	adoption_applications_reviewed_by_foreign ON DELETE SET NULL ON UPDATE RESTRICT	reviewed_by + Add column	paws_db	users	id
	Constraint name ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column	paws_db		
+ Add constraint					

Medical Records Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	medical_records_pet_id_foreign ON DELETE CASCADE ON UPDATE RESTRICT	pet_id + Add column	paws_db	pets	id
Drop	medical_records_vet_id_foreign ON DELETE RESTRICT ON UPDATE RESTRICT	vet_id + Add column	paws_db	users	id
	Constraint name ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column	paws_db		
+ Add constraint					

Vaccination Records Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	vaccination_records_admin	administered_by	paws_db	users	id
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			
Drop	vaccination_records_pet_id	pet_id	paws_db	pets	id
	ON DELETE CASCADE ON UPDATE RESTRICT	+ Add column			
	Constraint name		paws_db		
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			
+ Add constraint					

Payments Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	payments_adoption_applc	adoption_applicate	paws_db	adoption_applicate	id
	ON DELETE SET NULL ON UPDATE RESTRICT	+ Add column			
Drop	payments_payer_id_foreign	payer_id	paws_db	users	id
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			
Drop	payments_recorded_by_for	recorded_by	paws_db	users	id
	ON DELETE SET NULL ON UPDATE RESTRICT	+ Add column			
	Constraint name		paws_db		
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			
+ Add constraint					

Transaction Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	transactions_payment_id_fc	payment_id	paws_db	payments	id
	ON DELETE CASCADE ON UPDATE RESTRICT	+ Add column			
	Constraint name		paws_db		
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			
+ Add constraint					

Activity Logs Constraint

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	activity_logs_user_id_foreig	user_id	paws_db	users	id
	ON DELETE SET NULL ON UPDATE RESTRICT	+ Add column			
	Constraint name		paws_db		
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			
+ Add constraint					

System Notification Constraint

Actions	Constraint properties	Column 	Foreign key constraint (INNODB)		
			Database	Table	Column
 Drop	system_notifications_user_i	user_id	paws_db	users	id
	ON DELETE CASCADE ON UPDATE RESTRICT	+ Add column			
	Constraint name		paws_db		
	ON DELETE RESTRICT ON UPDATE RESTRICT	+ Add column			

[+ Add constraint](#)

VIII. Advance Database Features

8.1 Triggers

The following triggers are implemented in the PAWS database to automate critical operations and maintain data consistency.

Trigger 1: after_pet_approved_notify_staff

Its purpose is to automatically create a system notification for all the Staff users when a veterinarian marks a pet as fit for adoption.

Explanation: This trigger fires after an update on the PETS table, specifically when the vet_approved_by column changes from NULL to a non-null value, meaning a veterinarian has just approved the pet. Once triggered, it automatically inserts a new notification row into the SYSTEM_NOTIFICATIONS table for each active Staff user, alerting them that a pet has been medically cleared and is now ready for admin approval.

Details

Trigger name

after_pet_approved

Table

pets

Time

AFTER

Event

UPDATE

Definition

```
1 BEGIN
2   IF NEW.is_admin_approved = 1 AND OLD.is_admin_approved
   = 0 THEN
3     INSERT INTO system_notifications
4       (user_id, title, message, type, created_at,
        updated_at)
5       SELECT id, 'Pet Approved',
6         CONCAT(NEW.name, ' has been approved for adoption'),
7         'info', NOW(), NOW()
8       FROM users;
9   END IF;
10  END
```

Definer

root@localhost

Trigger 2: after_application_status_update_notify_adopter

Its purpose is to automatically notifies the adopter when their application status is changed by Staff or Admin.

Explanation: This trigger fires after an update on the ADOPTION_APPLICATIONS table, specifically when the status column value changes from one stage to another. Once triggered, it automatically inserts a new row into the ACTIVITY_LOGS table recording the old status, the new status, the user who made the change, and the timestamp, creating a complete audit trail of every application status update without requiring manual logging.

Details

Trigger name

after_application_status_ch

Table

adoption_applications

Time

AFTER

Event

UPDATE

Definition

```
1 BEGIN
2   IF NEW.status != OLD.status THEN
3     INSERT INTO activity_logs
4       (user_id, action, model_type, model_id,
5        description,
6         old_values, new_values, created_at, updated_at)
7     VALUES (
8       NEW.reviewed_by, 'updated',
9       'AdoptionApplication', NEW.id,
10      CONCAT('Application status changed to ',
11             NEW.status),
12      JSON_OBJECT('status', OLD.status),
13      JSON_OBJECT('status', NEW.status),
14      NOW(), NOW());
15   END IF;
```

Definer

root@localhost

Trigger 3: after_payment_recorded_update_application

Purpose: Automatically updates the payment status and generates a transaction log entry when a payment record is inserted.

Explanation: This trigger fires after every new record is inserted into the PAYMENTS table. Once triggered, it automatically inserts a corresponding row into the TRANSACTIONS table containing the payment reference, amount, gateway, method, status, and timestamp, creating a complete financial audit trail for every payment recorded in the system without requiring manual transaction entry.

Details

Trigger name

after_payment_inserted

Table

payments

Time

AFTER

Event

INSERT

Definition

```
1 BEGIN
2   INSERT INTO transactions
3     (payment_id, transaction_id, gateway,
4      amount, type, status, created_at, updated_at)
5   VALUES (
6     NEW.id,
7     CONCAT('TXN-', NEW.id),
8     'manual',
9     NEW.amount,
10    'payment',
11    NEW.status,
12    NOW(), NOW());
13 END
```

Definer

root@localhost

8.2 Views

View 1 - view_available_pets:

Its purpose is to provide a list of all pets available for adoption

SQL code:

```
Run SQL query/queries on table paws_db.VIEWS: ⓘ

1 CREATE OR REPLACE VIEW view_available_pets AS
2 SELECT
3   p.id,
4   p.name,
5   p.gender,
6   p.size,
7   p.color,
8   p.adoption_fee,
9   p.primary_image,
10  pc.name AS category_name,
11  b.name AS breed_name
12 FROM pets p
13 JOIN pet_categories pc ON p.pet_category_id = pc.id
14 JOIN breeds b ON p.breed_id = b.id
15 WHERE p.status = 'available'
16 AND p.is_admin_approved = 1
17 AND p.deleted_at IS NULL;
```

Browse Result:

Showing rows 0 - 0 (1 total, Query took 0.0019 seconds)

SELECT * FROM "view_available_pets"

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

	id	name	gender	size	color	adoption_fee	primary_image	category_name	breed_name
☐	9	Mimim number 3	female	small	calico	0.00	pets/yy7ftrKgyJ4zlw0eywFaP8LSqVxm6i2THbb0yCqH.jpg	Cat	Puspin (Pusa Pinoy)

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

View 2 - view_pet_details

Its purpose is to provides a complete pet profile

SQL code:

```
Run SQL query/queries on table paws_db.view_available_pets: ⓘ

1 CREATE OR REPLACE VIEW view_pet_details AS
2 SELECT
3     p.id,
4     p.name,
5     p.gender,
6     p.size,
7     p.color,
8     p.description,
9     p.is_neutered,
10    p.is_microchipped,
11    p.status,
12    p.adoption_fee,
13    p.primary_image,
14    p.is_admin_approved,
15    pc.name AS category_name,
16    b.name AS breed_name,
17    u.name AS added_by_name
18 FROM pets p
19 JOIN pet_categories pc ON p.pet_category_id = pc.id
20 JOIN breeds b ON p.breed_id = b.id
21 JOIN users u ON p.added_by = u.id
22 WHERE p.deleted_at IS NULL;
```

Browse Result:

Showing rows 0 - 7 (8 total, Query took 0.0008 seconds)

SELECT * FROM `view\_pet\_details`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Extra options

		id	name	gender	size	color	description	is_neutered	is_microchipped	status	adoption_fee	primary_image	is_admin_approved	category_name	breed_name	added_by_name	
☐				1	ilaga	male	medium and white	Brown and Very Lively Dog	0	0	adopted	2000.00	pets/7A9kx8fbEz5mSH84NyULZaFmE0MhsMarkCwXFT40.jpg	1	Dog	Aspin (Asong Pinoy)	Maria
☐				2	Tabudiong	male	small	black & white introvert dog	0	0	pending	1000.00	pets/NhClicvRH#thuwvu46e5GPoXq5ZQ2idalfq4pD.jpg	1	Dog	Aspin (Asong Pinoy)	Maria
☐				4	Tabudiong the 3rd	male	small	black & white dog he is a good	1	0	adopted	1000.00	pets/llPq7XOb6MJ8NA4sRiaUmHrpTXcabFvNP40eBwU.jpg	1	Dog	Aspin (Asong Pinoy)	Maria
☐				5	bulanday	female	medium	white she is a very good dog	0	0	adopted	2000.00	pets/zMeJ1vPgY8az7eewZ37Z7xRT7zqW8AQ8X5pCdEUI.jpg	1	Dog	Aspin (Asong Pinoy)	Maria
☐				6	Bulanda the 2nd	female	large	brown very underweight	0	0	adopted	2000.00	pets/AyHbGRwWJ8dCniuQAZmdrqrDXDE6zvsMbB3Gpq.jpg	1	Dog	Aspin (Asong Pinoy)	Maria
☐				7	Mimim number 1	female	small	tlapia cute ass hell	1	0	adopted	500.00	pets/luTL06P5bXR8pKpynJG7dDbwF2iG106Q45veEu.jpg	1	Cat	Puspin (Pusa Pinoy)	Maria
☐				8	Mimim number 2	female	small	tlapia very quirky	0	0	adopted	500.00	pets/B95cwMnSBN1jethGwhURIE46BcdHHEd10el8l.jpg	1	Cat	Puspin (Pusa Pinoy)	Maria
☐				9	Mimim number 3	female	small	calico Noisy cat	0	0	available	0.00	pets/y7ftrKgyJ4zW0eywFaP8LSqVxm6i2THbbdyCqH.jpg	1	Cat	Puspin (Pusa Pinoy)	Maria

View 3 - vw_application_summary

Its purpose is to provide a summary of all adoption applications with adopter and pet details.

SQL code:

```
Run SQL query/queries on table paws_db.view_pet_details: ⓘ

1 CREATE OR REPLACE VIEW vw_application_summary AS
2 SELECT
3     aa.id,
4     aa.application_number,
5     aa.status,
6     aa.submitted_at,
7     aa.reviewed_at,
8     p.name AS pet_name,
9     u1.name AS adopter_name,
10    u1.email AS adopter_email,
11    u2.name AS reviewed_by_name
12 FROM adoption_applications aa
13 JOIN pets p ON aa.pet_id = p.id
14 JOIN users u1 ON aa.adopter_id = u1.id
15 LEFT JOIN users u2 ON aa.reviewed_by = u2.id;
```

Browse Results:

Showing rows 0 - 9 (10 total, Query took 0.0011 seconds)

SELECT * FROM `vw\_application\_summary`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	id	application_number	status	submitted_at	reviewed_at	pet_name	adopter_name	adopter_email	reviewed_by_name
☐ Edit Copy Delete	2	PAWS-2026-00001	completed	2026-04-12 10:29:50	2026-04-12 11:02:37	Ilaga	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	3	PAWS-2026-00002	rejected	2026-04-12 16:22:19	2026-04-12 16:28:39	Tabudlong	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	4	PAWS-2026-00003	completed	2026-04-12 16:29:55	2026-04-16 05:36:41	Tabudlong	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	5	PAWS-2026-00004	completed	2026-04-12 16:39:26	2026-04-12 16:42:56	Tabudlong jr.	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	6	PAWS-2026-00005	completed	2026-04-12 16:44:19	2026-04-16 05:36:30	Tabudlong the 3rd	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	7	PAWS-2026-00006	completed	2026-04-12 16:52:50	2026-04-16 05:36:21	bulanday	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	8	PAWS-2026-00007	completed	2026-04-12 16:59:13	2026-04-16 05:36:05	Bulanda the 2nd	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	9	PAWS-2026-00008	completed	2026-04-15 05:58:55	2026-04-16 05:35:55	Miming number 1	Ana Cruz	adopter@paws.local	System Administrator
☐ Edit Copy Delete	10	PAWS-2026-00009	completed	2026-04-15 06:16:28	2026-04-15 06:17:54	Miming number 2	Angelo Tapzillog	Tapzillog@paws.com	System Administrator
☐ Edit Copy Delete	11	PAWS-2026-00010	completed	2026-04-16 13:25:31	2026-04-16 13:26:19	Macotin	Ana Cruz	adopter@paws.local	System Administrator

IX. System Features and Functionality

The PAWS system is organized to have eight functional modules that addresses a specific operational need of the shelter.

1. Role-Based Access Control (RBAC)

The system implements four user roles which are the Administrator, Staff, Veterinarian, and Adopter. Each role has a dedicated dashboard and access to only the features that is relevant to that role. Role assignments are managed through a many-to-many pivot table which allows a flexible multi role configuration and with a custom RoleMiddleware which enforces access restrictions at the route level.

2. Pet Profile Management

The Staff members can create and manage pet profiles that includes physical attributes like (name, gender, age, size, color), health flags (neutered, microchipped, special needs), adoption configuration (fee, fee type, admin approval status), and media (primary image and additional photos), and lastly, pets undergo a dual-approval process before becoming visible to adopters.

3. Adoption Application Workflow

Adopters can browse available pets and can submit a detailed adoption applications through an online form. Applications are moved through a defined status of progression from Submitted to Completed, with the Staff managing the reviews and the admin handling the final approval. Adopters can also track their application status in real time and withdraw applications that have not yet been approved.

4. Medical and Vaccination Records

Veterinarians have exclusive access in order to create medical records for each pet, documenting examination findings, diagnosis, treatment, and health status. Vaccination records capture vaccine name, manufacturer, batch number, administration date, and the next due date.

A fit for adoption flag on the medical record will triggers the notifications to Staff when a pet is medically cleared.

5. Payment Recording

Staff and Administrators can record payments linked to approved adoption applications. with each payment that is assigned to a unique reference number and supports a multiple payment method. Proof of payment files that can be uploaded and attached to the record. Every payment automatically generates a transaction log entry for auditing purposes.

6. Notification System

The system will generate a automatic system notifications for key events like for the adoption application submission, status changes, medical clearance, and payment recording. Each notification includes a title, message, type, icon, and optional action link. Users can mark notifications as read individually or all at once, and an unread badge count appears in the navigation sidebar.

7. Activity Logging

All of the major system actions are recorded within the activity log with the performing user, action name, affected model type and ID, description, old values, new values, IP address, and timestamp. This log provides a complete audit trail for administrators to review system activity and investigate discrepancies.

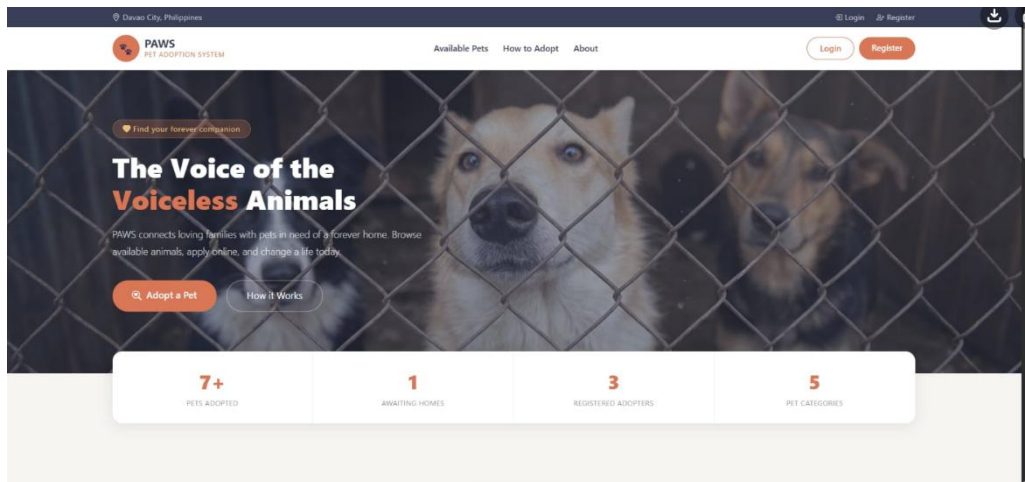
8. Reports and Analytics

The admin dashboard can provide a summary statistic that includes the total pets, available pets, pending applications, adopted pets, and the registered users. There is also a reports module that enables administrators to view adoption statistics, payment summaries, and activity trends, supporting informed decision making for shelter operations.

X. Prototype / User Interface

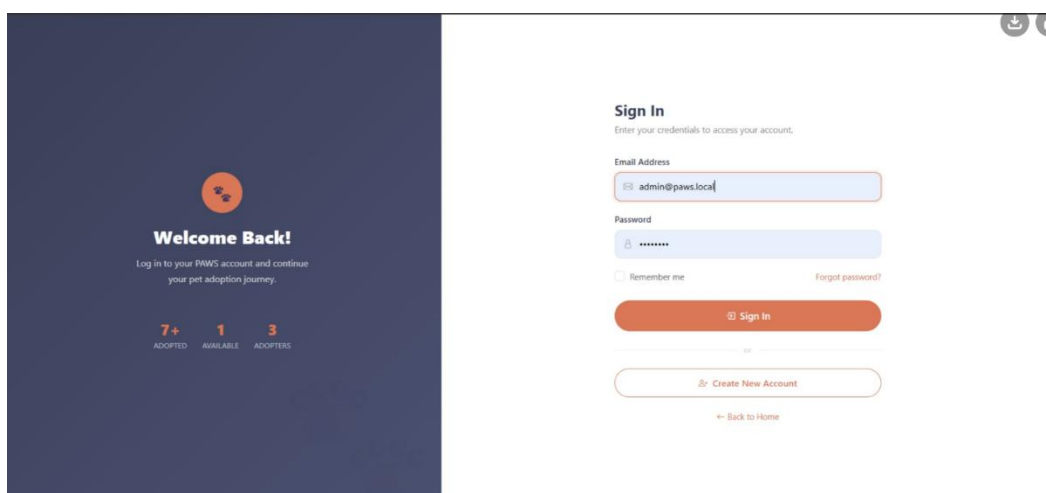
The following screenshots are illustrating the key interfaces of the PAWS system. With each screenshot that are accompanied by a brief description of the page and its function.

Landing Page



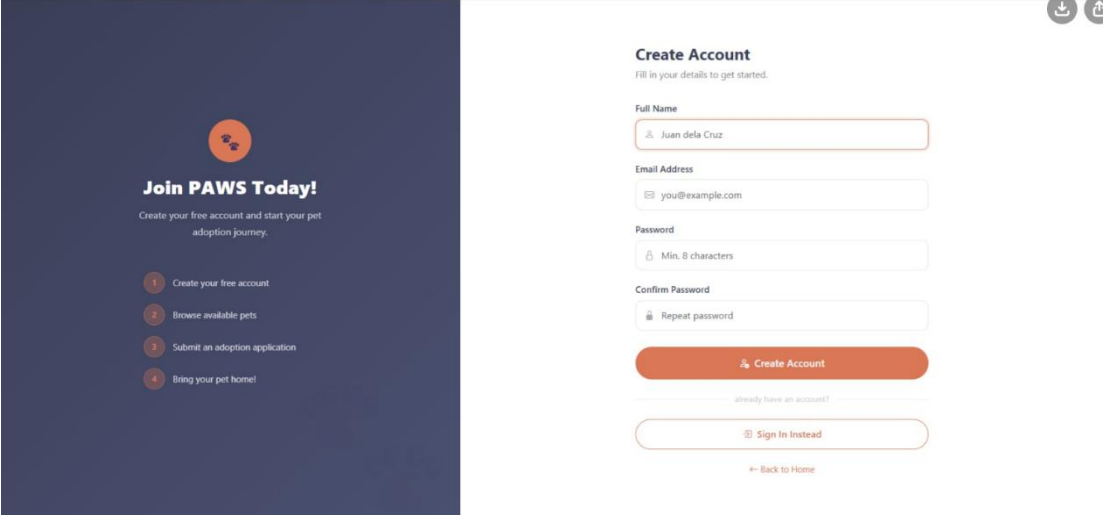
The landing page serves as the front page of the PAWS system. The landing page which provides the visitors with an overview of the shelter, highlights featured available pets, and encourages prospective adopters to create an account or log in. The page includes a navigation bar with links to the login and register pages, a hero section with a call-to-action button, and a section showcasing pets currently available for adoption.

Login Page



The login page allows the users to authenticate using their registered email address and password. A Forgot Password link is available for account recovery, and a Register link is provided for new adopter self-registration.

Register Page



The screenshot shows the 'Join PAWS Today!' registration page. On the left, a dark blue sidebar contains a paw print icon, the title 'Join PAWS Today!', a subtext 'Create your free account and start your pet adoption journey.', and a four-step process: 1. Create your free account, 2. Browse available pets, 3. Submit an adoption application, and 4. Bring your pet home! On the right, the 'Create Account' form is displayed. It includes fields for Full Name (filled with 'Juan dela Cruz'), Email Address (filled with 'you@example.com'), Password (with a 'Min. 8 characters' hint), and Confirm Password (with a 'Repeat password' hint). Below the fields are two buttons: 'Create Account' and 'Sign In Instead' (with a link 'already have an account?'). At the bottom is a 'Back to Home' link.

Join PAWS Today!
Create your free account and start your pet adoption journey.

- 1 Create your free account
- 2 Browse available pets
- 3 Submit an adoption application
- 4 Bring your pet home!

Create Account
Fill in your details to get started.

Full Name
Juan dela Cruz

Email Address
you@example.com

Password
Min. 8 characters

Confirm Password
Repeat password

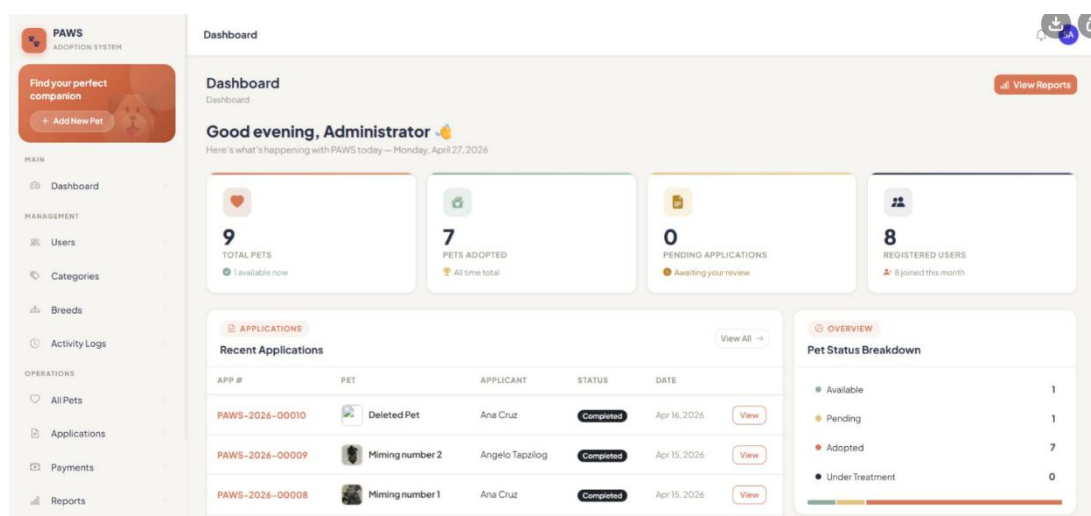
Create Account

already have an account?
Sign In Instead

Back to Home

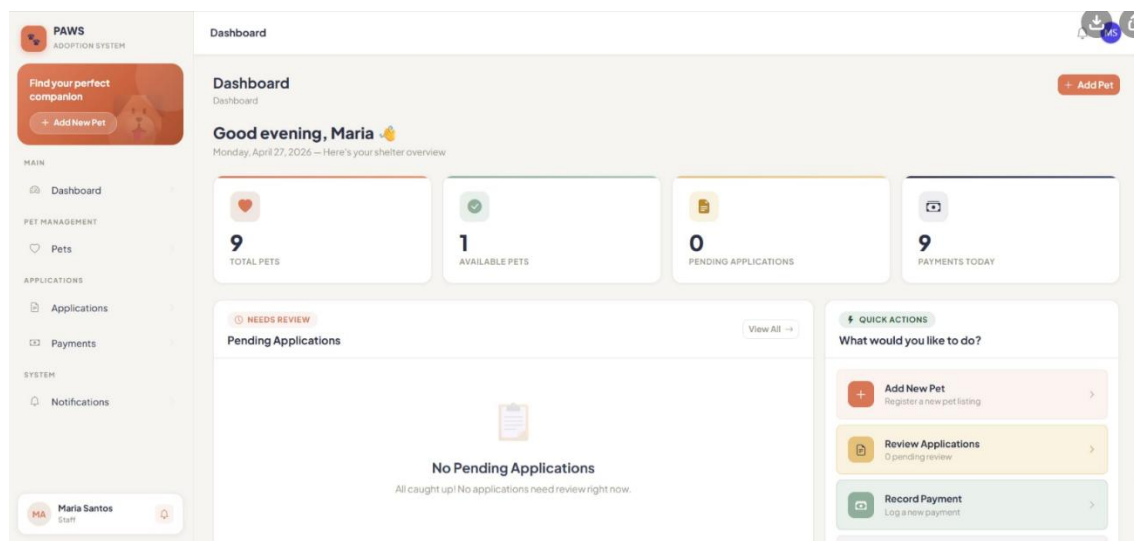
The new users of the system can register as an Adopter through the registration form, providing their name, email, and password. After a successful registration, the Adopter role is assigned automatically, and the user can be redirected to the Adopter Dashboard.

Admin Dashboard



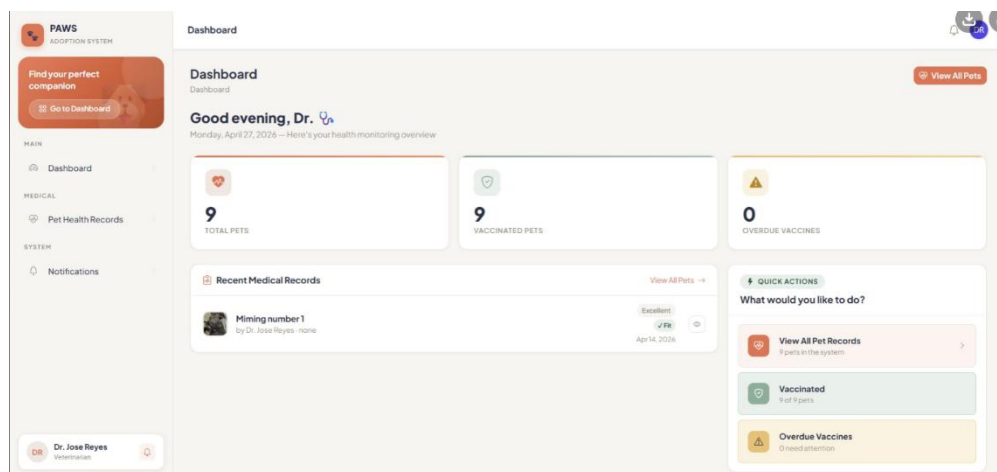
The Admin Dashboard is the one that displays the summary of the shelter statistics which includes the total pets, available pets, pending applications, total adoptions, and registered users. Below the stat cards, a table of recent applications and a list of pets pending admin approval are shown.

Staff Dashboard



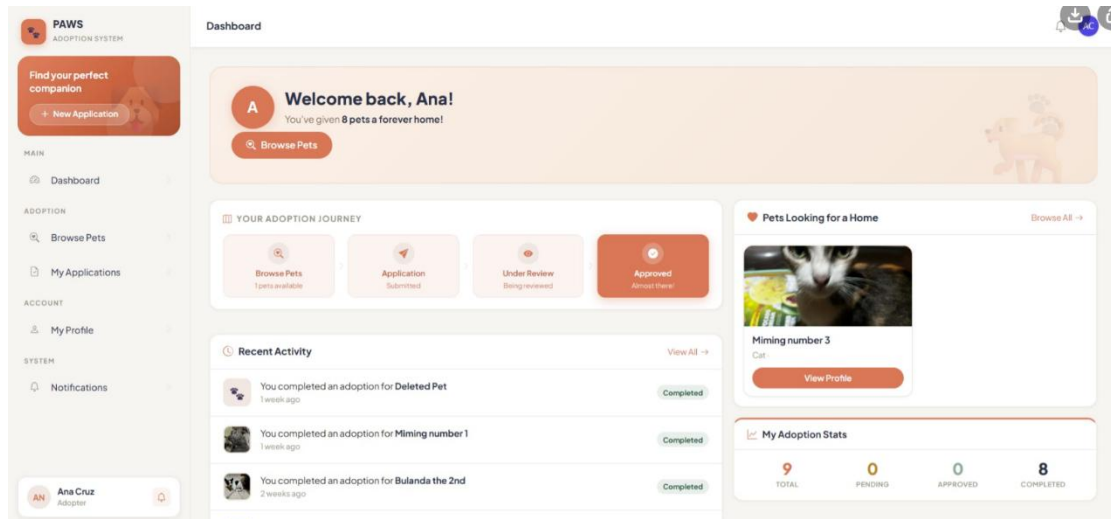
The Staff Dashboard shows operational statistics that are relevant to the day-to-day shelter management, which includes the pets in care, pending applications, and payments recorded. The sidebar provides quick access to pet management and application processing tools.

Vet Dashboard



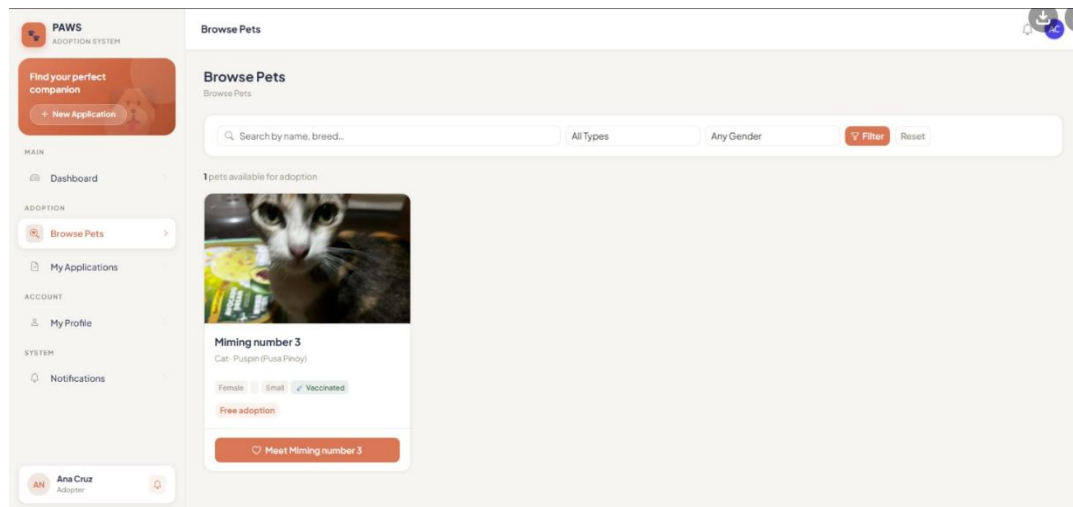
The Veterinarian Dashboard is the one that highlights the pets that requires an examination and displays a recent medical records and upcoming vaccination schedules. Vets can also navigate directly to medical record creation and vaccination logging from this view.

Adopter Dashboard



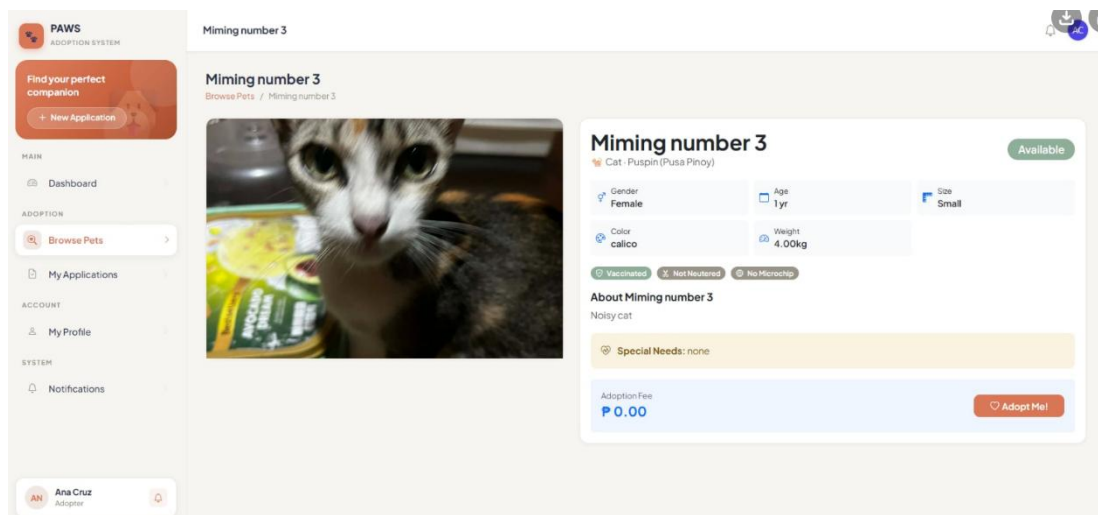
The Adopter Dashboard is the one that provides a personalized overview with a sidebar that leads to browse available pets, a count of submitted applications, and recent notification alerts. Adopters can track their application status from the My Applications section.

Browse Pets Page



The Browse Pets page displays pets that are ready for adoption in form of a grid layout. Adopters can filter by category and breed, and search by pet name. Each pet card shows the primary image, name, breed, age, gender, and adoption fee.

Pet Details Page



The Pet Detail page shows the full profile of a selected pet, which includes the physical attributes, medical clearance status, vaccination history, and the Adopt Me button is for eligible adopters.

Adoption Application Form

The screenshot displays the 'Adoption Application' form within the PAWS Adoption System. The interface includes a left sidebar with navigation links: MAIN (Dashboard), ADOPTION (Browse Pets, My Applications), ACCOUNT (My Profile), and SYSTEM (Notifications). The main content area is titled 'Adoption Application' and shows the progress 'Browse Pets / Miming number 3 / Apply'. The form is divided into two main sections: 'Personal Information' and 'Home & Lifestyle'. The 'Personal Information' section contains fields for Email Address (adopter@paws.local), Phone Number (+63 912 000 0004), Occupation (e.g. Teacher, Engineer), and Home Address (Davao City). The 'Home & Lifestyle' section includes a dropdown for 'Type of Housing', a text field for 'When you're away, who cares for the pet?' (e.g. Family member, pet sitter), and three checkboxes: 'I have a yard/garden', 'I have other pets', and 'I have children'. On the right side, there is a pet profile for 'Miming number 3', a Cat (Puspin/Pusa Pinoy), 1yr old, Female, with an adoption fee of P 0.00. A note at the bottom states: 'Processing typically takes 3-5 business days. We'll contact you for an interview.'

The adoption application form is the one that collects the applicant's personal information, home and lifestyle details, reason for adoption, emergency contact, and additional notes. The form is validated on submission, and the application is assigned the Submitted status upon successful completion.

Application Management (Staff/Admin)

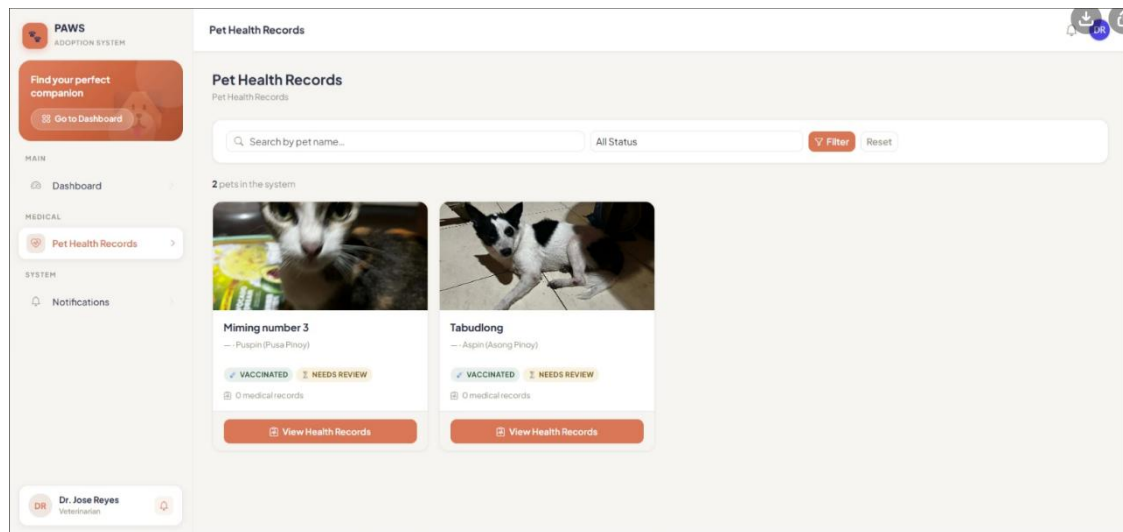
The top screenshot displays the 'Applications' management page. It features a sidebar with navigation options: MAIN (Dashboard), PET MANAGEMENT (Pets), APPLICATIONS (Applications, Payments), and SYSTEM (Notifications). The main content area shows a list of 10 applications with columns for APP #, PET, APPLICANT, STATUS, SUBMITTED, and ACTION. All applications shown have a status of 'Completed'.

APP #	PET	APPLICANT	STATUS	SUBMITTED	ACTION
PAWS-2026-00010	Pet Deleted	Ana Cruz adopter@paws.local	Completed	Apr 16, 2026	
PAWS-2026-00009	Miming number 2	Angelo Tapilog Tapilog@paws.com	Completed	Apr 15, 2026	
PAWS-2026-00008	Miming number 1	Ana Cruz adopter@paws.local	Completed	Apr 15, 2026	
PAWS-2026-00007	Bulanda the 2nd	Ana Cruz adopter@paws.local	Completed	Apr 12, 2026	
PAWS-2026-00006	bulanday	Ana Cruz adopter@paws.local	Completed	Apr 12, 2026	
PAWS-2026-00005	Tabulong the 3rd	Ana Cruz adopter@paws.local	Completed	Apr 12, 2026	

The bottom screenshot displays the 'Application Detail' page for application PAWS-2026-00010. It shows a 'Back' button, the application ID, and a 'Completed' status. The page is divided into two main sections: 'Pet Information' and 'Applicant Information'. The 'Applicant Information' section lists details for Ana Cruz, including email, phone, address, housing type, reason for adopting, experience with pets, and additional notes. A 'Timeline' section on the right shows the application submitted on Apr 16, 2026, at 01:25 PM and under review on Apr 16, 2026, at 01:26 PM.

The application management page is the one that lists all of the adoption applications with their status. Staff and Admin can filter by status, view full application details, update the status, schedule interviews, and record review notes from this interface.

Pet Health Records Page



The Pet Health Records page is accessible to veterinarians and is the one that displays all of the pets currently in the system. With each pet that is shown as a card with the pet's photo, name, breed, vaccination status, and medical record count. Veterinarians can search for a specific pet by name and filter by status. By clicking the View Health Records button on any pet card opens the full health history of that pet, allowing the veterinarian to add new medical records, vaccination entries, and update the pet's fitness for adoption status.

Medical Record Form

The screenshot shows the 'Add Medical Record' form in the PAWS Adoption System. The left sidebar is identical to the previous screenshot. The main content area is titled 'Add Medical Record' and contains a 'New Medical Record' form. The form fields include: 'Examination Date *' (27/04/2026), 'Health Status *' (Select Status), 'Fit for Adoption?' (radio buttons for 'Yes, fit for adoption' and 'No'), 'Diagnosis' (text input), 'Treatment' (text input), 'Next Checkup Date' (text input with format dd/mm/yyyy), and 'Notes' (text input). The bottom of the sidebar shows the user 'Dr. Jose Reyes' as a Veterinarian.

The medical record form allows the veterinarians to document the pet's examination findings, including symptoms, diagnosis, treatment, weight, health status, and fitness for adoption.

Vaccination Record Form

The screenshot shows the 'Add Vaccination Record' form in the PAWS Adoption System. The left sidebar contains navigation links: PAWS ADOPTION SYSTEM, Find your perfect companion (Go to Dashboard), MAIN (Dashboard), MEDICAL (Pet Health Records), and SYSTEM (Notifications). The main content area is titled 'Add Vaccination Record' and includes a breadcrumb trail: Pets / Miming number 3 / Add Vaccination. A pet profile card for 'Miming number 3' (Cat) is shown with a 'Vaccinated' status. The 'Vaccination Details' form includes fields for Vaccine Name* (e.g., Rabies, DHPP, Bordetella...), Vaccine Batch / Lot # (Optional lot number), Date Administered* (27/04/2026), Next Due Date (dd/mm/yyyy), Manufacturer (e.g., Zoetis, Merck...), and Dosage / Route (e.g., 1mL subcutaneous). There is a text area for Notes (Any observations or additional notes...) and a checkbox for 'Mark pet as vaccinated after saving'. At the bottom right are 'Cancel' and 'Save Vaccination' buttons.

The vaccination record form is the one that captures the vaccine name, manufacturer, batch number, date administered, next due date, and booster status for each vaccination entry linked to a specific pet.

Notification Page

The screenshot shows the 'Notifications' page in the PAWS Adoption System. The left sidebar is identical to the previous screenshot. The main content area is titled 'Notifications' and includes a search bar for 'All Notifications' and a 'Mark All Read' button. The notification list contains eight entries, all with a green checkmark icon and the title 'Pet Listing Approved'. The messages are: 'Your listing for Maccotin has been approved and is now public.', 'Your listing for Miming number 3 has been approved and is now public.', 'Your listing for Miming number 2 has been approved and is now public.', 'Your listing for Miming number 1 has been approved and is now public.', 'Your listing for Bulanda the 2nd has been approved and is now public.', 'Your listing for bulanday has been approved and is now public.', and 'Your listing for Tabudong the 3rd has been approved and is now public.'. The timestamps for the notifications are: 1 week ago, 1 week ago, 1 week ago, 2 weeks ago, 2 weeks ago, 2 weeks ago, and 2 weeks ago.

The notifications page lists all system-generated alerts for the pet approval, with icons, timestamps, and action links. Users can mark individual notifications as read or clear all notifications at once.

XI. Testing and Results

The PAWS system was tested using a functional testing in order to verify that each of the module operates correctly according to its defined requirements. Testing was conducted using the four test accounts: admin@paws.local, staff@paws.local, vet@paws.local, and adopter@paws.local, all with the password 'password'.

Test Case	Module	Expected Result	Actual Result	Status
TC-01	Login	Valid credentials redirect to correct role dashboard	Redirected correctly for all 4 roles	PASS
TC-02	Login	Invalid credentials show error message	Error message displayed; no redirect	PASS
TC-03	Pet Management	Staff can add a pet with image upload	Pet created and image stored in storage	PASS
TC-04	Pet Management	Pet is not visible to adopters without admin approval	Pet hidden from Browse Pets page	PASS
TC-05	Medical Records	Vet can mark pet as fit for adoption	fit_for_adoption set to true; notification sent to Staff	PASS
TC-06	Adoption Application	Adopter can submit application for available pet	Application created with Submitted status	PASS
TC-07	Adoption Workflow	Staff can move application to Under Review	Status updated; adopter notified	PASS
TC-08	Adoption Workflow	Admin can approve application	Status set to Approved; pet status set to Pending	PASS
TC-09	Payment	Staff can record payment linked to application	Payment created with unique reference number	PASS

Test Case	Module	Expected Result	Actual Result	Status
TC-10	Notifications	Adopter receives notification on status change	Notification appears in adopter notification panel	PASS
TC-11	Activity Log	All actions are recorded in activity log	Entries created for pet creation, application update, payment	PASS
TC-12	RBAC	Adopter cannot access Staff/Admin routes	403 Forbidden page displayed	PASS

XII. Conclusion and Recommendations

Conclusion

The PAWS: Pet Adoption Transaction Management System has successfully addressed the operational challenges that are faced by the animal shelters that are still using the manual and paper-based processes. The system centralizes a pet profiles, medical records, adoption applications, payment records, and user management into a single, accessible web platform which was built on Laravel 12, Bootstrap 5, and MySQL.

It is possible through a role-based access control, which has four distinct user roles which are the Administrator, Staff, Veterinarian, and Adopter with each of them to interact with the system through tailored interfaces and permission sets. The approval process for pet listings ensures that only medically cleared and administratively approved pets are visible to adopters, raising the standard of animal welfare within the platform.

The structured adoption application workflow, with a combined system notifications and a complete activity audit log, will ensure the user that every stage of the adoption process is documented, traceable, and transparent. The payment module with unique reference numbers and transaction logging provides financial accountability that was previously absent in manual systems.

Overall, the PAWS system has demonstrated about how a well-structured relational database combined with a modern web framework can significantly improve the efficiency, accuracy, and transparency of a animal shelter operations.

Recommendations

The following enhancements are recommended for future development of the PAWS system:

1. Integrate an online payment gateway such as PayMongo or Stripe to allow adopters to pay adoption fees directly through the system without requiring manual payment recording by staff.
2. Develop a mobile responsive version or dedicated mobile application to improve accessibility for adopters who primarily use smartphones.
3. Add automated email notifications using Laravel's Mail system or a third-party service such as Mailgun or Resend to complement in-system notifications with external alerts.
4. Implement a real-time chat or messaging feature between adopters, staff, and administrators to reduce reliance on external communication channel.
5. Expand the reporting module in order to include the exportable PDF and Excel reports for adoption statistics, payment summaries, and medical record histories.

XIII. References

- [1] Hettiarachchi, C. (2019). Automated Animal Shelter Management System for Animal Welfare Groups of Sri Lanka. *International Journal for Research in Applied Science and Engineering Technology*, 7, 957-964. <https://doi.org/10.22214/ijraset.2019.12153>.
- [2] Zhang, T. (2020). Advancing Non-profit Organizations Through Technology and Informatization An analysis on improving the underdeveloped animal shelters and rescues industry. *E3S Web of Conferences*. <https://doi.org/10.1051/e3sconf/202021801047>.
- [3] Horecka, K., & Neal, S. (2022). Critical Problems for Research in Animal Sheltering, a Conceptual Analysis. *Frontiers in Veterinary Science*, 9. <https://doi.org/10.3389/fvets.2022.804154>.
- [4] Ahire, N., Binnar, T., Bodke, S., Sagar, S., & Patil, J. (2025). A Smart and Integrated Animal Rescue Application for Efficient Incident Reporting. *International Journal of Advanced Research in Science, Communication and Technology*. <https://doi.org/10.48175/ijarsct-24636>.